

Run Bundle Generator for pi2/NRStitcher

This Streamlit application generates ready-to-execute "Run Bundles" for the [pi2/NRStitcher](#) pipeline. It simplifies the complex process of creating configuration files and execution scripts for both local workstations and Slurm-managed clusters (like Misha).

Features

- **Interactive Configuration:** Easily input dataset parameters (dimensions, overlap, voxel size) via a GUI.
- **Auto-Detection:** Automatically detects your dataset dimensions from file metadata.
- **Smart Script Generation:** Creates `run_local.bat` (Windows), `run_local.sh` (Linux/Mac), and `run_nrstitcher.sbatch` (Slurm) with intelligent backend detection.
- **Tiles View:** (Optional) Creates a `tiles/` folder with symlinks, renaming your files to a structured format (`tile_{t}_z_{z}_c_{c}.tif`) expected by some viewers, without duplicating data.
- **Preview:** Visually inspect your tiles and verify coordinate mapping before generating.
- **Slurm Resource Recommendation:** Estimates required Partition, CPU, Memory, and Time based on your dataset size.

Quick Start

1. Get the Code & Stitcher

You'll need this UI application and the core stitcher ([pi2](#)).

A. The Run Bundle App (This UI) Clone this repository to your machine:

```
git clone https://github.com/allison714/nrstitcher-preprocessing.git
cd nrstitcher-preprocessing
```

B. The Core Stitcher ([pi2](#))

- **Windows (Recommended):** Download the pre-compiled binary distribution ([pi2-v4.5-win-no-openc1](#)). The app automatically supports it if placed at `D:\pi2-v4.5-win-no-openc1` (or you can link it manually). You do NOT need to clone the pi2 repo.
- **Source Building:** If you must build from source, clone the [pi2](#) repo (`git clone https://github.com/arttumieltinen/pi2.git`).

2. Getting Started

There are two ways to start the application, depending on your operating system:

Option A: Windows (Easiest) An automated batch script is provided that will automatically find your Conda installation, create the required `stitch_app` environment, install dependencies, and launch the UI.

1. Simply double-click `RUN_ME.bat` in the project folder!

Option B: Mac / Linux (Manual Anaconda Prompt) If you are on Mac/Linux, or prefer to set up the environment manually:

1. Open your terminal (or **Anaconda Prompt**).
2. Navigate to the downloaded project directory: `cd path/to/nrstitcher-preprocessing`
3. Create and activate the conda environment, then install dependencies:

```
conda create -n stitch_app python=3.9 -y
conda activate stitch_app
pip install -r requirements.txt
```

4. Run the application:

```
streamlit run app.py
```

The application will open in your default web browser automatically.

Workflow

1. **Select Data:** Input your dimensional parameters, overlapping preferences, and physical resolutions. !
[Data Configuration](file:///C:/Users/allis/.gemini/antigravity/brain/a36c68f1-6232-4658-8d75-fabd2bf012b9/screenshot_inputs.png)
2. **Verify:** Use the "**Preview Tiles**" section to visually check if your coordinate mappings follow your intended scan path.
3. **Configure Execution:**
 - **Target Environment:** Choose "Local Workstation" or "Misha Cluster (Slurm)". ![Execution Setup](file:///C:/Users/allis/.gemini/antigravity/brain/a36c68f1-6232-4658-8d75-fabd2bf012b9/screenshot_tooltip.png)
 - **Backend config:** The app attempts to auto-detect `pi2` or `nrstitcher`. You can override this if needed.
4. **Generate:** Click "Generate Run Bundle".

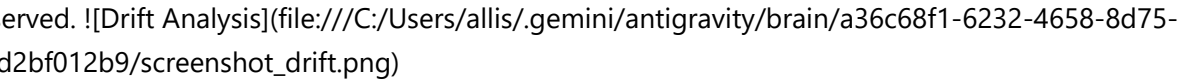
Post-Generation Quality Control (QC)

After generating and running your stitched bundle, point the application to your output directory to compute advanced quality control analytics.

Warping Diagnostics

Identifies mechanical issues by isolating non-linear 3D spatial drags across tile boundaries. Pinpoints exact regions of problematic physical topologies using robust median density filters. ![Warping Diagnostics](file:///C:/Users/allis/.gemini/antigravity/brain/a36c68f1-6232-4658-8d75-fabd2bf012b9/screenshot_warping.png)

Intensity Drift Analysis

Mathematically decouples baseline illumination decay and laser-power inconsistencies from actual physiological density measurements. This allows for safe, robust detection of photobleaching trajectories and automatically scaffolds a physical structural file rewrite (Gain Correction) if unacceptable attenuation is observed.  (file:///C:/Users/allis/.gemini/antigravity/brain/a36c68f1-6232-4658-8d75-fabd2bf012b9/screenshot_drift.png)

Output

The app creates a new folder (e.g., `MyDataset_local` or `MyDataset_slurm`) containing:

- `stitch_settings.txt`: The coordinate configuration for the stitcher.
- `dataset_manifest.json`: A record of your settings for reproducibility.
- `run_local.bat` / `run_local.sh` / `run_nrstitcher.sbatch`: The specific script to **run the actual stitching**.
- `tiles/`: (Optional) The symlinked view of your data.

Local Workstation Execution

Prerequisite: `pi2` / `nrstitcher`

For local stitching, you need the `pi2` software. Since this is not a public PyPI package, you have two options:

1. Portable Bundle (Recommended):

- Download the [pi2 source code](#).
- In the App, under **Local Config**, paste the path to your `pi2` folder in "**Path to 'pi2' Package Source**".
- The App will **embed** a copy of `pi2` into the run bundle (`tools/pi2`).
- The generated script will automatically use this embedded copy, meaning you don't need to install it in your environment!

2. Auto-Download from GitHub:

- If you leave the "Path to 'pi2'" blank, the generated script will attempt to:
 1. Create a fresh Conda environment (`stitch_app`) if it doesn't exist.
 2. Install dependencies (`numpy`, `tifffile`, `scikit-image`).
 3. Run `pip install git+https://github.com/arttumienninen/pi2`.
- **Requires:** `git` must be installed and available in your command prompt.

Running the Stitcher

1. Navigate to the generated bundle folder.
2. Double-click `run_local.bat` (Windows) or run `./run_local.sh` (Linux/Mac).
3. The script will:
 - Activate the `stitch_app` environment.
 - Run the stitch command.
 - (Optional) Convert the output to Neuroglancer Precomputed format if selected.

Neuroglancer Visualization

If you selected **Neuroglancer Precomputed** output, you can instantly view your 3D stitched volume in your browser:

1. Open **Anaconda Prompt**.
2. Navigate to your stitched output folder (e.g., `cd D:\StitchScratch\260224_local...`).
3. Run the included server script:

```
python serve.py
```

4. Open Chrome and go to neuroglancer-demo.appspot.com.
5. Click the **+** icon in the top left and add your local Source:
`precomputed://http://localhost:8000/precomputed.`

Output

The app creates a new folder containing:

- `stitch_settings.txt`: Configuration file.
- `dataset_manifest.json`: JSON record of settings.
- `run_local.bat` / `run_local.sh`: Intelligent execution scripts.
- `serve.py`: Local web server for Neuroglancer visualization.
- `tools/`: (If using Portable Bundle) Contains the embedded `pi2` package.

Troubleshooting

Auto-Install Failed

- **Error:** `pip install git+... failed`
- **Cause:** You likely don't have `git` installed, or you are behind a firewall.
- **Fix:** Download `pi2` manually from GitHub, extract it, and use the **"Path to 'pi2' Package Source"** field in the App to create a Portable Bundle instead.

"Conda not found"

- The script tries to find `conda` automatically. If it fails, open the App and check **"Local Config > Conda Init Script"**. Ensure it points to your actual `conda.bat` (usually `C:\Users\Username\anaconda3\condabin\conda.bat`).

Exporting this README to PDF

If you use Visual Studio Code and wish to save this documentation as a PDF, you can automate it using the `Markdown PDF` extension:

1. Install the `Markdown PDF` extension by `yzane`.
2. Add the `"markdown-pdf.convertOnSave": true` option to your VS Code `settings.json`.
3. Restart Visual Studio Code.

4. Open this Markdown file and save it (Ctrl+S / Cmd+S). The PDF will auto-generate in the same folder.